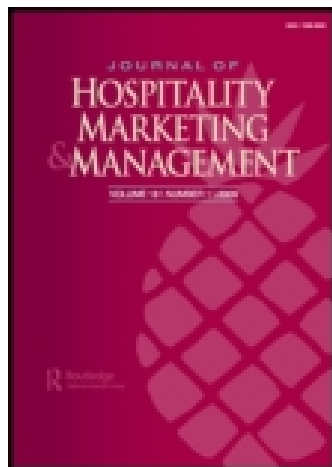


This article was downloaded by: [University of Florida]

On: 15 April 2015, At: 08:33

Publisher: Routledge

Informa Ltd Registered in England and Wales Registered Number: 1072954 Registered office: Mortimer House, 37-41 Mortimer Street, London W1T 3JH, UK



[Click for updates](#)

Journal of Hospitality Marketing & Management

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/whmm20>

User-Generated Content as a Research Mode in Tourism and Hospitality Applications: Topics, Methods, and Software

Weilin Lu^a & Svetlana Stepchenkova^a

^a Department of Tourism, Recreation, and Sport Management, College of Health and Human Performance, University of Florida, Gainesville, Florida, USA

Accepted author version posted online: 28 Apr 2014. Published online: 14 Nov 2014.

To cite this article: Weilin Lu & Svetlana Stepchenkova (2015) User-Generated Content as a Research Mode in Tourism and Hospitality Applications: Topics, Methods, and Software, Journal of Hospitality Marketing & Management, 24:2, 119-154, DOI: [10.1080/19368623.2014.907758](https://doi.org/10.1080/19368623.2014.907758)

To link to this article: <http://dx.doi.org/10.1080/19368623.2014.907758>

PLEASE SCROLL DOWN FOR ARTICLE

Taylor & Francis makes every effort to ensure the accuracy of all the information (the "Content") contained in the publications on our platform. However, Taylor & Francis, our agents, and our licensors make no representations or warranties whatsoever as to the accuracy, completeness, or suitability for any purpose of the Content. Any opinions and views expressed in this publication are the opinions and views of the authors, and are not the views of or endorsed by Taylor & Francis. The accuracy of the Content should not be relied upon and should be independently verified with primary sources of information. Taylor and Francis shall not be liable for any losses, actions, claims, proceedings, demands, costs, expenses, damages, and other liabilities whatsoever or howsoever caused arising directly or indirectly in connection with, in relation to or arising out of the use of the Content.

This article may be used for research, teaching, and private study purposes. Any substantial or systematic reproduction, redistribution, reselling, loan, sub-licensing, systematic supply, or distribution in any form to anyone is expressly forbidden. Terms &

User-Generated Content as a Research Mode in Tourism and Hospitality Applications: Topics, Methods, and Software

WEILIN LU and SVETLANA STEPCHENKOVA

Department of Tourism, Recreation, and Sport Management, College of Health and Human Performance, University of Florida, Gainesville, Florida, USA

The rapid growth of information generated by consumers of tourism and hospitality services calls for a systematic review of how user-generated content (UGC) has been applied in tourism and hospitality research. This study surveyed 122 peer-reviewed journal articles and conference proceedings that used UGC as a data source. The study investigates (a) the scope of the tourism and hospitality issues that are addressed using available UGC; (b) the methods that have been applied to UGC data to achieve research objectives; and (c) the software that has been used to collect UGC and extract information from large UGC data sets. The study also presents the emerging topics and challenges in UGC research.

KEYWORDS *consumer-generated media, hospitality, social media, systematic review, tourism, user-generated content, user-generated media*

INTRODUCTION

The amount of digital data accumulated from countries and industries doubles every 1.2 years (KnowWPC, 2012):

Address correspondence to Svetlana Stepchenkova, University of Florida, College of Health and Human Performance, Eric Friedheim Tourism Institute, Department of Tourism, Recreation, and Sport Management, PO Box 118208, Gainesville, FL 32611-8208, USA. E-mail: svetlana.step@ufl.edu

Color versions of one or more of the figures in the article can be found online at <http://www.tandfonline.com/whmm>.

Scholars are clamoring for access to the massive quantities of information produced by and about people, things, and their interactions. Diverse groups argue about the potential benefits and costs of analyzing genetic sequences, social media interactions, health records, phone logs, government records, and other digital traces left by people. (Boyd & Crawford, 2012, p. 663)

This voluminous information is often referred to as “big data,” and user-generated content (UGC) is a part of it. The explosion of UGC is a result of the Internet and mobile technologies, which provide individuals with unprecedented power to instantaneously add “digital traces” when performing tasks such as reviewing airline, hotel, and restaurant services, lodging a customer complaint, documenting a travel experience, or uploading photos and videos to the global big data bank. Facebook, Flickr, TripAdvisor, Twitter, Yelp, and other social media platforms contain huge volumes of UGC that are suitable for research with respect to tourism and hospitality applications.

UGC has been described as creative work that is published on publicly accessible websites and is created without a direct link to monetary profit or commercial interest (Organisation for Economic Co-operation and Development, 2007). Since its appearance in the 1990s (Shao, 2009), UGC websites have evolved into multiple forms: virtual communities (e.g., LonelyPlanet), consumer reviews (e.g., Yelp), personal stand-alone blogs, blog aggregators (e.g., LiveJournal), and microblogging platforms (e.g., Tumblr); as well as social networks (e.g., Facebook), media sharing tools (e.g., Flickr, YouTube), and wikis (e.g., Wikitravel; Xiang & Gretzel, 2010). Arguably, blogs and virtual communities are the earliest UGC providers. A blog is “a small Website, usually maintained by one person, that is updated on a regular basis and has a high concentration of repeat visitors” Barrett (1999, p. 1). A new form of blog (i.e., the microblog) is composed of “short, succinct messages, used by both consumers and businesses to share news, post status updates, and carry on conversations” (Tse & Zhang, 2013, p. 316). Virtual communities and, to a large extent, social networks are “social aggregations that emerge from the Net when enough people carry on public discussions long enough . . . to form webs of personal relationships in cyberspace” (Rheingold, 1993, pp. 57-58). Among a plethora of UGC types, wikis represent dynamically changing UGC; they are websites that allow users to edit site content, hence, operating on a principle of collaborative trust (Goodchild, 2007). Finally, consumer review websites provide platforms for broadcasting information about consumer experiences for the benefit of future users of products and services, while media-sharing sites allow exchange of photos and videos.

UGC is considered an electronic form of word-of-mouth marketing (eWOM; J. Kim & Hardin, 2010; J. Q. Zhang, Craciun, & Shin, 2010) that has traditionally been defined as the extent to which a customer informs

friends, relatives, and colleagues of an event that has created a certain level of satisfaction (Soderlund, 1998). UGC is widely used by consumers of hospitality and tourism services both for information sharing and as an information source for making decisions: online consumer accounts are perceived as more likely to contain up-to-date, informative and reliable information that is rich in detail and highly relevant (Gretzel & Yoo, 2008). In a survey of Generation Y, the largest consumer group in the United States, eight out of 10 respondents said that UGC that originates from people they do not know influences their buying decisions (Bazaar Voice, 2012). On the supplier side, the perceived trustworthiness of UGC (Cox, Burgess, Sellitto, & Buultjens, 2009) renders it as a viable channel for businesses to monitor consumer feedback and preferences, to communicate with current and potential customers, and to use UGC as a source of information for organizational change or brand co-creation (Tussyadiah & Zach, 2013).

Compared to traditional surveys, UGC has a number of advantages as a research mode, including data availability, speed and simplicity of data collection, and its use as a nonintrusive source that lacks the complications associated with direct interactions with human subjects. However, the distribution and sheer amount of UGC data available present methodological challenges in collecting, organizing, and analyzing the bulk of this material in a quantifiable, time-efficient, and ethical manner. In addition, the question of whom UGC represents, and what the boundaries for generalizability of results are, is still open, as gathering demographic and psychographic information of UGC creators has been problematic from both the data availability and ethics standpoints (Johnson, Sieber, Magnien, & Ariwi, 2012). Researchers have also noted, "Big Data is less about data that is big than it is about a capacity to search, aggregate, and cross-reference large data sets" (Boyd & Crawford, 2012, p. 663). Hence, while researchers and industry practitioners have taken advantage of the power of UGC to create value for destinations, business operations, and industry consumers, studies have documented the difficulty that exists in the location, extraction, and interpretation of UGC, a component that may limit the potential implementation of UGC in research (Akehurst, 2009; Carson, 2008).

The aforementioned features and limitations of UGC may naturally lead to specific areas of applicability of UGC for tourism and hospitality research, while various types of UGC data (e.g., textual or visual) may be conducive for particular approaches of data collection and analysis. Given volumes of UGC that is potentially available for research, the current situation with utilizing computer software in UGC studies also calls for investigation. Reviews of prior research assist in analyzing historical patterns, predicting future trends, and guiding the integration of research findings into business practice (D. Leung, Law, van Hoof, & Buhalis, 2013). To date, there have been several studies that systematically surveyed previous cross-sectoral research from the tourism and hospitality, social media, and information science fields (e.g., N. Li, Buhalis, & Zhang, 2013; Hjalager & Nordin, 2011; Leung et al., 2013;

Racherla & King, 2012). However, the only review that focused specifically on UGC is by Banyai and Glover (2011), who investigated methods of analyzing written content in blogs.

Thus, the current study was conducted to examine how UGC data have been used in empirical tourism and hospitality research and what are the issues and problems pertinent to the industry that have been addressed by the use of UGC data. Specifically, three research objectives were formulated: (a) to analyze the scope of tourism and hospitality issues that have been addressed using UGC as a research mode; (b) to examine methods and analytical procedures that were applied to UGC data; and (c) to survey the software that has been used to collect and extract information from UGC data sets. The current review is considered a reference guide for researchers and practitioners who wish to identify topics of scholarly inquiry, current methodology practices, and technological developments concerning travel- and hospitality-related UGC. The study also presents the emerging topics and challenges in UGC research.

METHOD

Data Collection

The authors applied a systematic review approach to appraise a cumulative outcome of empirical research that used UGC as a data source to address issues in tourism and hospitality applications. A systematic review is “a review in which there is a comprehensive search for relevant studies on a specific topic, and those identified are then appraised and synthesized according to a pre-determined explicit method” (Klassen, Jadad, & Moher, 1998, p. 701). Google Scholar is regarded as the “first port of call for researchers,” providing users with a convenient, rapid, and free way to gather relevant literature (SAGE, n.d.); it returns articles, conference proceedings, theses, books, abstracts and other documents from scholarly publishers, professional societies, university libraries, and web sites, in the order of the frequency that links to resources are being clicked on (Maslov & Redner, 2008; van Aalst, 2010). However, Google Scholar was criticized for duplication, inaccurate citation counts, and other search issues (Jasco, 2008). Thus, the authors also searched Web of Science, and online archives of the “top five” tourism and “top five” hospitality journals (McKercher, Law, & Lam, 2006) to include the most relevant and comprehensive collection of academic articles on the topic. To minimize the effect of *false negatives*, i.e., relevant articles that are not found, multiple search terms were used, including, but not limited to, “user-generated content,” “user-generated media,” “consumer-generated media,” “social media,” “social network sites,” “blogs,” “online reviews,” “e-complaints,” and “electronic word-of-mouth” in various combinations with words such as “tourism,” “travel,” “hospitality,” “hotel,” and

“restaurant.” For each search term combination, the first 50 results returned by Google Scholar and all results returned by Web of Science were evaluated for inclusion in the study.

The key to systematic review is that the criteria for the inclusion or exclusion of studies in the review is explicit from the outset, and while others may not agree with the inclusions, the criteria for such inclusions, and thus the scope of the review, are clearly delimited. (Weed, 2006, p. 6)

In order to avoid the effect of *false positives* (i.e., articles that had only a tangential relevance to the focus of this review), and obtain a homogeneous data set in terms of relevance, quality of scholarly work, accessibility, language, and format (Miles & Huberman, 1994), several inclusion/exclusion criteria were formulated. Thus, only English-language academic articles from journals that adhere to a double-blind review process and peer-reviewed conference articles were selected. Only empirical research that had a well-defined UGC sample in the form of textual, pictorial, video, or numerical entries (e.g., rating scores) was considered; thus theoretical/conceptual articles on the nature of UGC or review studies were not included. Articles that used travel-related UGC to demonstrate an idea or answer a question from a field of study other than tourism and hospitality (e.g., Ye, Zhang, & Law [2009] compared the efficiency of three supervised machine learning algorithms by classifying sentiments in online travel reviews) were not included. Articles published in both conference proceedings and academic journals were included only once, unless the authors, research purpose, or research design had been changed; the older or preliminary version of the same article was excluded.

Considering the still-emerging status of the UGC subject area, no start date for publications was set. The data collection concluded in April 2013; thus, articles that appeared after April 2013 were not included. Researchers speed-read each article's abstract, introduction, methodology, and as much text from relevant sections as needed (Crawford-Welch & McCleary, 1992; Stepchenkova & Mills, 2010) to evaluate the article suitability for this review according to the criteria. A total of 577 articles were speed-read, and the final sample contained 122 articles. The earliest published research was a study of online complaints against an airline by Harrison-Walker (2001) that appeared in the *Journal of Services Marketing*. There has since been a steady growth of publications that used UGC as a research instrument (Figure 1). Overall, articles from the final sample were published in 75 tourism and hospitality journals (Jamal, Smith, & Watson, 2008; McKercher et al., 2006; Pechlaner, Zehrer, Matzler, & Abfalter, 2004), 10 nontourism and hospitality journals, and 37 conference proceedings. Among journals, the *Annals of Tourism Research* and the *Journal of Hospitality Marketing & Management* had the largest number of publications (8), and the proceedings of the *Information*

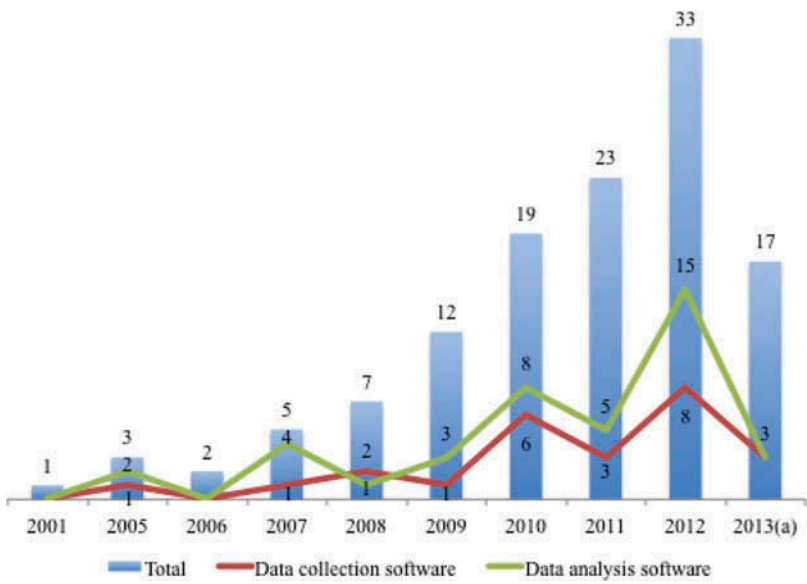


FIGURE 1 Number of publications by year and software usage. (a) Between January 1, 2013 and April 30, 2013.

and Communication Technologies in Tourism (ENTER) conference was the outlet for 16 articles.

Data Analysis

The selected 122 articles were systematically surveyed using an approach that included classification of the articles into a number of relevant categories and then synthesis of the articles along these categories. The classification phase involved three flows of analytical activity: data reduction, data display, and verification of the data, with data collection and data reduction not being detached, disconnected stages but rather a continuous process of data analysis (Miles & Huberman, 1994; Y. Wang & Fesenmaier, 2007). Categories were predetermined by objectives of this review and included: tourism and hospitality setting, source of UGC data, type of UGC data, sample size, topic area, analytical procedure, and specialized software. The classification process resulted in the following values for the categories:

- Tourism and hospitality setting: Destination (54); lodging (48); restaurant (8); airline (4); cruise (2); attraction (1); and online travel agent (1), with several studies conducted in more than one setting.
- Source of UGC data: Consumer review websites (63); blogs (33); media-sharing websites (20); social networks (11); virtual communities (8); and wikis (1), with several studies using more than one source of UGC data.

Studies that used data from Twitter, which can be considered as both microblogging and social network platform (Twitter, n.d.), were classified under the blogs category.

- Type of UGC data: This category has the values of texts, photos, videos, and rating scores. Texts were the dominant data type, and a number of articles used data of more than one type.
- Sample size: Across the corpus of surveyed studies, sample size differs dramatically, from 13 blogs (Banyai, 2012) to more than 20 million textual units and photos (X. Lu, Wang, Yang, Pang, & Zhang, 2010).
- Topic area: Service quality (50), destination image and reputation (26), UGC as e WOM (25), experiences and behavior (14), and mobility patterns (7). The authors aimed to produce topic areas that were broad enough to include more than a few studies, yet distinctive enough to allow for as clear a delineation as possible between different topic areas. Another consideration was to end up with a “manageable” number of topic areas to allow for a convenient summary of the articles. Keywords supplied by the authors of the respective articles were also consulted to verify classification. Few cases, where the authors somewhat struggled with article attribution, were resolved in a discussion.
- Analytical procedure: These are listed in Table 1 using terminology given by the authors of the respective studies.
- Specialized software: It was classified into software instruments used for data collection (25), data analysis (41), and both data collection and analysis (19).

RESULTS AND DISCUSSION

What Is the Scope of Tourism and Hospitality Issues That Have Been Addressed Using UGC Data?

Service Quality. Articles in this area are largely concerned with the issues of customer satisfaction, complaint behavior, as well as service failure and recovery. Most studies used customer reviews from TripAdvisor and other similar websites as their UGC source. The majority of studies were conducted within the lodging setting, with just a few studies conducted in the airline, online travel agent, and restaurant settings. Service quality studies provide useful feedback for industry managers to understand what aspects of their products and services are valued by customers the most (e.g., H. Li, Ye, & Law, 2012; Magnini, Crotts, & Zehrer, 2011) and, subsequently, how to distribute more effectively limited resources under their command to increase customer satisfaction. As a research mode, UGC seems well suited for this purpose, as it does not predetermine the issues of importance for consumers nor restrict them in expressing their opinions.

TABLE 1 Corpus of surveyed articles.

Author (year)	Tourism and hospitality setting	Source of UGC data	Type of UGC data (sample size)	Topic area	Analytical procedure	Specialized software
Harrison-Walker (2001)	Airline	Consumer review	T (551)	S	Content analysis	
Axup & Viller (2005)	Destination	Blog/Media sharing	T/P (6,720)	E	Content analysis	
Kurashima, Tezuka, & Tanaka (2005)	Destination	Blog	T (50)	M	Language processing, geo-visualization	DC (Crawler); DA (Chasen morphological analyzer)
C. C. Lee & Hu (2005)	Lodging	Consumer review	T (222)	S	Content analysis, text mining, cluster analysis	DA (Catpac)
Y. S. Lin & Huang (2006)	Destination	Media sharing	T (301)	eW	Content analysis	
Z. Lu & Feng (2006)	Lodging	Consumer review	T (200)	S	Content analysis, correlation analysis	
Choi, Lehto, & Morrison (2007)	Destination	Blog	T/P (n/a)	I	Text mining, content analysis, correspondence analysis	DA (Catpac)
Girardin, Dal Fiore, Blat, & Ratti (2007)	Destination	Media sharing	P (n/a)	M	Geo-visualization	DC/DA (SDA)
Lehto, Park, Park, & Lehto (2007)	Online travel agent	Consumer review	T (672)	S	Text mining, expert judgment	DA (Text Miner)
Pan, MacLaurin, & Grotts (2007)	Destination	Blog	T (40)	I	Content analysis, thematic network analysis	DA (TextAnalyst/Nvivo)
Woodside, Cruickshank, & Dehuang (2007)	Destination	Blog	T/P (n/a)	I	Story mapping	

Berger & Greenspan (2008)	Destination	Blog	T (n/a)	E	Narrative analysis	
Carson (2008)	Destination	Blog	T (76)	E	Content analysis	
Girardin, Clabrese, Dal Fiore, Ratti, & Blat (2008)	Destination	Media sharing	P (n/a)	M	Geo-visualization	DC/DA (SDA)
Jeong & Jeon (2008)	Lodging	Consumer review	RS (n/a)	S	Analysis of means (parametric), linear regression	
Watson, Morgan, & Hemmington (2008)	Restaurant	Blog	T (n/a)	eW	Discourse analysis	
Wenger (2008)	Destination	Blog	T (188)	I	Content analysis	
Ye, Gu, Chen, & Law (2008)	Lodging	Consumer review	T (52,641)	S	Regression analysis	DC (Crawler)
Au, Buhalis, & Law (2009)	Lodging	Consumer review	T (453)	S	Content analysis	
Black & Kelly (2009)	Lodging	Consumer review	T (429)	eW	Content analysis, analysis of means (parametric)	
Bosangit, McCabe, & Hibbert (2009)	Destination	Blog	T (30)	eW	Narrative analysis	
Grotts, Mason, & Davis (2009)	Lodging	Consumer review	T (n/a)	S	Content analysis, stance-shift analysis	
Dwivedi (2009)	Destination	Virtual community	T (100)	I	Word frequency	
Dwivedi, Yadav, & Patel (2009)	Destination	Virtual community	T (n/a)	I	Word frequency	
Otto, Sanford, & Chuang (2009)	Lodging	Consumer review	RS (n/a)	S	Correlation, linear regression analysis	
Popescu, Grefenstette, & Moellic (2009)	Destination	Media sharing	P (723,303)	M	Geo-visualization	DC (Flickr API); DA (texture-color descriptor)
Schmallegger & Carson (2009)	Destination	Consumer review/Blog	T (27)	I	Content analysis	
Tussyadiah & Fesenmaier (2009)	Destination	Media sharing	V/T (696)	eW	Content analysis	DA (Catpac)

(Continued)

TABLE 1 (Continued)

Author (year)	Tourism and hospitality setting	Source of UGC data	Type of UGC data (sample size)	Topic area	Analytical procedure	Specialized software
Yoo & Gretzel (2009)	Lodging	Consumer review	T (82)	eW	Content analysis	DA (General Inquirer)
T. Zheng, Youn, & Kincaid (2009)	Lodging	Consumer review	T (504)	S	Content analysis	
Arsal, Woosnam, Baldwin, & Backman (2010)	Destination	Virtual community	T (74)	eW	Thematic network analysis	
Au, Law, & Buhalis (2010)	Lodging	Consumer review	T (964)	S	Content analysis, analysis of means (parametric)	
Banyai (2010)	Destination	Blog	T (20)	I	Content analysis	
Enoch & Grossman (2010)	Destination	Blog	T (29)	E	Content analysis	
García-Barriocanal, Sicilia, & Korfiatis (2010)	Lodging	Consumer review	T (412)	S	Language processing	DC (Crawler)
Gu & Ye (2010)	Lodging	Consumer review	RS (402,750)	S	Linear regression	DC (Crawler)
Inversini, Marchiori, Dedekind, & Cantoni (2010)	Lodging	Wiki/Consumer review/Media sharing/Blog/Virtual community	T (73)	I	Content analysis	
Jiang, Gretzel, & Law (2010)	Lodging	Consumer review	T (407)	S	Content analysis, linear regression	DA (General Inquirer)
Kline, Barreda, Gerdes, & Gregory (2010)	Lodging	Consumer review	T (360)	S	Word frequency	DC (Web spider)/DA (ManyEyes)
Law & Cheung (2010)	Destination	Blog	T (120)	I	Content analysis	DC (Crawler); DA (SDA)
X. Lu, Wang, Yang, Pang, & Zhang (2010)	Destination	Blog/Media sharing	T/P (20 million+)	M	Geo-visualization	

O'Connor (2010)	Lodging	Consumer review	T (500)	S	Content analysis	DA (NVivo7)
Pantelidis (2010)	Restaurant	Consumer review	T (2,471)	S	Content analysis	
Pengiran-Kahar, Syed-Ahmad, Ismail, & Murphy (2010)	Destination	Media sharing	P (589)	I	Content analysis	
Sparks & Browning (2010)	Lodging	Consumer review	T (200)	S	Content analysis, narrative analysis	DA (NVivo8)
Stringam & Gerdes (2010)	Lodging	Consumer review	T/RS (60,000+)	S	Word frequency	DC (Web spider); DA (SDA)
Stringam, Gerdes, & Vanleeuwen (2010)	Lodging	Consumer review	RS (60,000+)	S	Correlation analysis	DC (Web spider/SDA)
Tuzovic (2010)	Airline	Consumer review	T (141)	S	Content analysis	DA (MAXQDA 2007)
Z. Zhang, Ye, Law, & Li (2010)	Restaurant	Consumer review	RS (n/a)	S	Regression analysis	
Aggett (2011)	Cruise	Consumer review	T (241)	S	Content analysis	
Bulchand-Gidumal, Melián-González, & López-Valcárcel (2011)	Lodging	Consumer review	RS (n/a)	S	Linear regression	
Capriello, Mason, Davis, & Crotts (2011)	Destination	Consumer review	T (800)	E	Content analysis, semantic analysis, stance-shift analysis	
Ip, Cheung, Law & Au (2011)	Destination	Consumer review	T (595)	I	Content analysis	
Jani & Hwang (2011)	Destination	Virtual community	T (214)	I	Content analysis	DA (TextSTAT)
Jani, Jeong, & Hwang (2011)	Destination	Virtual community	T (220)	eW	Content analysis, chi-square test, t-test	
H. Lee, Law, & Murphy (2011)	Lodging	Consumer review	RS (928)	eW	Correlation, ANOVA	
D. Leung, Law, & Lee (2011)	Destination	Consumer review	T (2,247)	I	Content analysis	
X. Li & Wang (2011)	Destination	Blog	T (89)	I	Word frequency, content analysis	DA (Atlas.ti)

(Continued)

TABLE 1 (Continued)

Author (year)	Tourism and hospitality setting	Source of UGC data	Type of UGC data (sample size)	Topic area	Analytical procedure	Specialized software
W. Lu & Stepchenkova (2011)	Lodging	Consumer review	T (373)	S	Content analysis, analysis of means (non-parametric) Content analysis	
Maansson (2011)	Destination	Social network/Media sharing/Blog	T/P (n/a)	eW		
Magnini, Grotts, & Zehrer (2011)	Lodging	Blog	T (743)	S	Text mining, content analysis, chi-square test	
Marchiori, Eynard, Inversini, Cantoni, & Cerretti (2011)	Lodging	Consumer review	T (22,064)	eW	Statistical comparison	DC (Web scraper)
Maurer & Schaich (2011)	Lodging	Consumer review	T (352)	S	Content analysis, ANOVA, linear regression	
O'Connor (2011)	Lodging	Social network	T/P/V (n/a)	eW	Content analysis	
Pang, Hao, Yuan, Hu, Cai, & Zhang (2011)	Destination	Blog/Media sharing	T/P (116,000)	I	Text mining, photo-matching	DC (Crawler); DA (YahooPlacement)
Paris (2011)	Various	Blog/Media sharing/Social network	T (60+)	eW	Content analysis	
Reino & Hay (2011)	Destination	Media sharing	V (320)	eW	Content analysis	
Son (2011)	Destination	Blog	T (15)	I	Content analysis, word frequency	DA (Nvivo)

Stepchenkova & Zhan (2013)	Destination	Media sharing	P (500)	I	Content analysis, chi-square analysis, attribute co-occurrences analysis	
Ye, Law, Gu, & Chen (2011)	Lodging	Consumer review	RS (40,424)	S	Regression analysis	DC (Crawler)
Zehrer, Grotts, & Magnini (2011)	Lodging	Consumer review	T (134)	eW	Analysis of means (non-parametric)	
Z. Zhang, Ye, & Law (2011)	Lodging	Consumer review	RS (n/a)	S	Regression analysis	
Ardizzone, Di Miceli, La Cascia, & Mazzola (2012)	Destination	Media sharing	P (3,010)	M	Geo-visualization	DC (Flickr API); DA (Google Maps API)
Azariah (2012)	Destination	Blog	T/P (n/a)	eW	Discourse analysis	
Banyai (2012)	Destination	Blog	T/P/N (13)	I	Content analysis	DA (Catpac)
Chaves, Gomes, & Pedron (2012)	Lodging	Consumer review	T/RS (1,500)	S	Content analysis	
Chen, Chen, & Wang (2012)	Destination	Blog	T (232)	I	Content analysis, chi-square analysis	DA (CAQDAS)
Grotts, Davis, & Mason (2012)	Lodging	Consumer review	T (200)	E	Semantic analysis, stance-shift analysis	DA (Corpus based software)
Del Chiappa & Dall'Aglio (2012)	Lodging	Consumer review	T/RS (731)	S	Content analysis, analysis of means (parametric)	DA (UAM Corpus Tool)
Ekiz, Khoo-Lattimore, & Memarzadeh (2012)	Lodging	Consumer review	T (320)	S	Thematic analysis	
Giannakos, Pateli, & Pappas (2012)	Lodging	Consumer review	T (1,000)	S	Content analysis	
Hays, Page, & Buhalis (2012)	Destination	Social network	T/P (n/a)	eW	Content analysis	
Hvass & Munar (2012)	Airline	Social network	T/P (164,165)	eW	Content analysis	

(Continued)

TABLE 1 (Continued)

Author (year)	Tourism and hospitality setting	Source of UGC data	Type of UGC data (sample size)	Topic area	Analytical procedure	Specialized software
S. E. Kim & Ichto (2012)	Various	Consumer review	T (316)	S	Content analysis, correspondence analysis	
G. Li, Law, Vu, & Rong (2012)	Lodging	Consumer review	RS (5,443)	S	Choquet Integral	DC (Visual Web Ripper); DA (SDA)
H. Li, Ye, & Law (2012)	Lodging	Consumer review	T (42,668)	S	Text mining, content analysis, cluster analysis	DC (SDA); DA (ICTCLAS 2011)
Lim, Chung, & Weaver (2012)	Destination	Media sharing	T (17,798)	I	Content analysis	DA (Catpac)
Y. Liu (2012)	Restaurant	Social network	T/P/V (n/a)	eW	Content analysis	
S. Liu, Law, Rong, Li, & Hall (2013)	Lodging	Consumer review	T/RS (6,196)	S	Sentiment mining	DA (SENTIWORDNET 3.0)
Q. Lu, Xiao, & Ye (2012)	Lodging	Consumer review	RS (284,851)	S	Regression analysis	DC (Crawler, LocoySpider)
Magnini, Kara, Crotts, & Zehrer (2012)	Lodging	Blog	T (334)	S	Text mining, chi-square test	
Murakami, Kawamura, & Suzuki (2012b)	Destination	Blog	T (886)	eW	Text mining (morphological analysis and concordance analysis)	DA (TinyTextMinder, AntCone)
Murakami, Kawamura, & Suzuki (2012a)	Destination	Blog	T (1,410)	eW	Content analysis	
Nimrod (2012)	Destination	Virtual community	T (3,425)	eW	Content analysis, thematic analysis	DC/DA (Forum Monitoring System)
Ougut & OnurTacs (2012)	Lodging	Consumer review	RS (950)	S	Log-linear regression analysis	

Onge & du Gros (2012)	Destination	Blog	T (n/a)	E	Thematic analysis	DA (Nvivo)
Pearce (2012)	Destination	Blog	T/P (297)	E	Content analysis	
Shakeela & Weaver (2012)	Destination	Blog/Media sharing/Social network	T (1,454)	E	Discourse analysis	
Sigala (2012)	Restaurant	Virtual community	T (n/a)	E	Content analysis	
Stylianou-Lambert (2012)	Destination	Media sharing	P (400)	I	Content analysis	
Tse & Zhang (2012)	Destination	Blog	T (600) P (3,751) T (202)	I	Content analysis	
Wang, Park, & Fesenmaier (2012)	Various	Consumer review	T (202)	E	Content analysis	DC (SDA)
Ye, Li, Wang, & Law (2012)	Lodging	Consumer review	RS (43,726)	S	Factor analysis, regression analysis	DC (SDA)
J. J. Zhang & Mao (2012)	Lodging	Consumer review	T (3,197) RS (4,141)	S	Content analysis, regression analysis, ANOVA	DA (Catpac)
Y.-T. Zheng, Zha, & Chua (2012)	Destination	Media sharing	P (769,000)	M	Geo-visualization, topological analysis of travel route	DC (Flickr API); DA (SDA)
Goulding, Saren, & Lindridge (2013)	Attraction	Blog	T (n/a)	E	Netnographic technique	
De Ascaniis, Bischof, & Cantoni (2013)	Destination	Consumer review	T (500)	I	Corpus analysis	
Duan, Cao, Yu, & Levy (2013)	Lodging	Consumer review	T/RS (70,103)	S	Sentiment analysis (language processing), linear regression	
Duverger (2013)	Lodging	Consumer review	RS (n/a)	S	Linear generalized method of moments (GMM)	DC (Certain social media monitoring tools); DA (Natural Language Toolkit [NLTK])
Kwok & Yu (2013)	Restaurant	Social network	T/P/V (982)	eW	Text mining, t-test	DC (Facebook Query Language [FQL]); DA (SVM, light software package)

(Continued)

TABLE 1 (Continued)

Author (year)	Tourism and hospitality setting	Source of UGC data	Type of UGC data (sample size)	Topic area	Analytical procedure	Specialized software
R. Leung, Schuckert, & Yeung (2013)	Airline	Social network	T (184)	eW	Content analysis	
Levy, Duan, & Boo (2013)	Lodging	Consumer review	T (2,171)	S	Content analysis, chi-square test	
Mkono (2013)	Restaurant	Consumer review	T (678)	E	Thematic analysis	
Mkono, Markwell, & Wilson (2013)	Various	Consumer review	T (285)	E	Thematic analysis	
Paris & Rubin (2013)	Destination	Social network	T (n/a)	eW	Case analysis	
Park & Allen (2013)	Lodging	Consumer review	T (n/a)	S	Case analysis	
Ramanatha & Ramanathan (2013)	Lodging	Consumer review	RS (24,544)	S	Multivariate regression analysis, path analysis	
Sari, Alikılıç, & Onat (2013)	Lodging	Consumer review	T (397)	S	Content analysis, chi-square test	
Stepchenkova, Kirilenko, & Kim (2013)	Destination	Social network	T (114,732)	I	Content analysis, polarity analysis	DC (Twitter API/SDA); DA (Catpac/SDA)
Syed-Ahmad, Musa, Klobas, & Murphy (2013)	Destination	Media sharing	P (589) T (320)	I	Content analysis, t-test	
Theng Loo, Boo, & Khoo-Lattimore (2013)	Restaurant	Consumer review	T (1,032)	S	Content analysis	
Zhang, Ye, Song, & Liu (2013)	Cruise	Consumer review	RS (44,993)	S	Regression analysis	

Note. Various = lodging, airline, land transportation, car rental, cruise, restaurant, destination, e-travel and/or attraction; P = photo; RS = ratings; T = text; V = video; E = experiences and behavior; eW = UGC as eWOM; I = destination image and reputation; M = mobility patterns; S = service quality; DC = data collection process; DA = data analysis process; SDA = self-developed application.

The surveyed studies cited employee rudeness, employee incompetence, and misinformation as primary reasons for online complaints (e.g., Sparks & Browning, 2010). Despite evidence that the presence of managerial response could substantially increase hotel sales (Q. Ye et al., 2008), service providers have not yet sufficiently integrated managerial response to online complaints into their daily practices (C. C. Lee & Hu, 2005; O'Connor, 2010). The lodging industry is particularly vulnerable to the negative effect of customer dissatisfaction expressed online because a substantial share of hotel booking transactions occur online (Q. Ye, Law, Gu, & Chen, 2011). Posting complaints and negative reviews is largely motivated by the inability of service providers to adequately resolve the initial complaints that are generally conveyed through hotel staff. Hotel managers should thus seriously consider using management strategies such as apology, promptness in response, explanation, attentiveness, redress, and effort as a means of service recovery (Au, Buhalis, & Law, 2009).

Destination Image and Reputation. For more than four decades, research on destination image produced hundreds of studies; these studies primarily used human subjects for data collection (Pike, 2002). As this review indicates, UGC is now considered an alternative mode for monitoring destination image and reputation. UGC is not only a low-cost channel for obtaining data, but it provides up-to-date feedback on destination performance from thousands of actual and potential users of the destination product. While “positioning a destination’s image . . . is a core activity of destination marketing organizations (DMOs), including national and state tourist offices, around the world” (Day, Skidmore, & Koller, 2002, p. 177), with ever-increasing UGC outreach, positioning is “not so much what you say about your products or company as much as it is what your customers say about you” (McKenna, 1991, p. 44). Therefore, monitoring existing customer perceptions and experiences that are “broadcasted” in blogs, virtual communities, and social networks for subsequent managerial action is more important than ever.

The destination image and reputation topic area has several orientations. One is to explore the perceived images of various destinations by specific marketing segments. For example, the two visitor segments to Flinder Range, Australia, namely, four-wheel drivers and young international travelers, highlighted different destination characteristics in their travel blogs (Schmallegger & Carson, 2009). Another orientation is a comparative analysis of destination images as projected by UGC creators and various travel intermediaries, government sources, and general media. Such analyses consistently captured differences in destination images transmitted via various information channels (e.g., Choi, Lehto, & Morrison, 2007; Dwivedi, Yadav, & Patel, 2009; Law & Cheung, 2010). Additionally, an orientation exists that examines positive and negative sentiments towards a destination, its strengths and weaknesses as perceived by tourists, and the comparative favorability of online and

offline images (Law & Cheung, 2010; Wenger, 2008). Finally, from a longitudinal perspective, researchers assessed how a destination's reputation develops online and classified the reputation drivers (Inversini, Marchiori, Dedekind, & Cantoni, 2010), as well as examined the similarities and differences between earlier and current destination representations (D. Leung, Law, & Lee, 2011).

UGC as eWOM. The main focus of the eWOM stream of studies is the issue of UGC influence on travel-related decisions and trust in UGC information. Research investigated the authoritative power of various information sources on decision making of potential tourists (e.g., Arsal, Woosnam, Baldwin, & Backman, 2010). Thus, residents were found to be more influential with respect to on-site decisions, such as those related to food and beverage and to those concerning safety, whereas experienced travelers were more influential with respect to decisions concerning accommodations, transportation, and travel costs. The studies also found that reviews documenting effective service recovery were perceived as the most helpful (Black & Kelley, 2009) and demonstrated that congruency in multiple evaluations was helpful as well (Zehrer, Crotts, & Magnini, 2011). However, the issue of truthfulness of UGC information and, consequently, the validity of the UGC data exists. Since the online deception has always been a concern, studies identified characteristics of fraudulent UGC by comparing language structure of deceptive and truthful hotel reviews (Yoo & Gretzel, 2009). With respect to travel blogs, Zehrer et al. (2011) observed that while fraudulent postings may exist, the methods to identify such cases are crude (23.5% of all postings in their study were suspect) and likely lead to false positives.

While aforementioned research looked at persuading power of online content generated by consumers, in the recent years studies appeared that focused on the eWOM power of UGC produced by suppliers of tourism and hospitality products (Hays, Page, & Buhalis, 2012; Hvass & Munar, 2012; Kwok & Yu, 2013; R. Leung, Schuckert, & Yeung, 2013; Y. Liu, 2012; O'Connor, 2011). Facebook, in particular, proved itself as conducive to this type of research. Thus, Kwok and Yu (2013) investigated what types of social media messages work best in business-to-consumer communications in restaurant setting; they found that conversational messages received higher endorsement from Facebook users than marketing communications. However, while many hospitality firms are present in social media, their level of activity and engagement with consumers is low and the use of this communication channel is often ineffective (O'Connor, 2011, with respect to hotel chains). This finding was echoed in the airline setting as well: researchers found that Facebook pages were mostly used for delivery of promotional materials, reaching to new customers, information distribution, as well as inviting customer engagement, rather than for publishing user-relevant material and responding to comments of current and potential customers (R. Leung et al., 2013).

Experiences and Behavior. UGC, and specifically tourist blogs, have proven to be a near-perfect source for researchers to identify emerging niche markets and document new trends in tourist experiences and behavior. Thus, Pan, MacLaurin, and Crotts (2007, p. 19) analyzed 40 blogs that described the travel experiences of visitors to Charleston, South Carolina, USA, and concluded, “travel blogs qualitatively cover every aspect of a visitor’s trip.” Axup and Viller (2005) were concerned with the way in which backpackers communicate and trade advice while on the road. Their study showed that backpackers, represented by young travelers who use iPhones, iPads, and other technological tools, are most likely to enthusiastically share their adventures with their “online-but-connected” peers. In another study, the interconnectedness of mountain adventure tourists and their local hosts in the Kumbu region of Nepal was explored using a Mount Everest expedition blog (Berger & Greenspan, 2008). An emerging topic in UGC research is studying human behavior in real time. For example, Paris and Rubin (2013) used Facebook and Twitter posts by backpacker tourists caught in the Chilean earthquake of 2010 and in the street protests in Bangkok, Thailand, in 2006 to follow the crisis-induced behavior through various stages of crisis development and resolution. The crisis-induced behavior is the area of study where the UGC mode wins over the traditional survey approach, and research using Twitter data seems the most promising in this regard.

Mobility Patterns. The wide geographical distribution of mobile devices, particularly smart-phones, allows posting messages and/or photos from virtually anywhere in the world and enables individuals to act as “sensors” with respect to global events (Goodchild, 2007). The review detected a growing interest in understanding the tourist mobility patterns, i.e., determining and locating events and attractions at a destination, identifying the most visited locations, and reconstructing tourist routes. A small but distinctive group of studies investigated the spatial and temporal mobility patterns by tracking “digital footprints” of visitors through geo-referenced photos and cell phone network data (Ardizzone, Di Miceli, La Cascia, & Mazzola, 2012; Girardin, Dal Fiore, Blat, & Ratti, 2007; Girardin, Calabrese, Fiore, Ratti, & Blat, 2008; Kurashima, Tezuka, & Tanaka, 2005; X. Lu et al., 2010). Specifically, Girardin and colleagues sampled geo-tagged photos on Flickr that were posted by 3,348 tourists in Florence, Italy, over a 2-year period. By tracking the digital footprints of tourists, the researchers not only captured tourist movements and popular activities and attractions, but they also provided practical insight for urban planners, traffic engineers, and tourism authorities on the density of tourists in the area. While these studies were not set out to explain tourist motivation or behavior, they did provide graphical visualization of tourist behavior that is of use to various destination stakeholders.

What Methods and Analytical Procedures Have Been Used to Analyze UGC Data?

In conducting a systematic review, the accepted practice is to classify research methods as quantitative, qualitative, or mixed-methods. However, the authors found that in dealing with research that utilizes UGC data, this classification is rarely clear-cut nor it is informative. There were just a few studies that could have been classified as purely quantitative or qualitative. For example, in the surveyed corpus, 68 studies used content analysis, the technique that has quantitative (as it results in obtaining a frequency matrix) and qualitative (as it involves a high degree of reasoning, interpretation, and expert judgment) elements; hence, it was considered a mixed-method procedure by the *Handbook of Research Methods in Tourism* (Dwyer, Gill, & Seetaram, 2013). The authors could not help to notice, however, that there were two distinctly different groups of studies in the surveyed corpus: those that do not utilize the big data aspect of UGC and those that do utilize it. The first group typically works with limited sample sizes that are manageable for manual data handling and applies some form of content analysis to textual data, sometimes, if the rating scores are involved, in combination with a quantitative procedure such as a regression. Such an approach, generally, allows for finely granulated details of analysis. The other group of studies uses more technologically advanced approaches that involve application of specialized software to collect UGC data and management of the resulting data sets. The big data studies introduce the latest developments in information science methods into tourism and hospitality research, which allows for large sample sizes and the integration of various software instruments. The results of such analyses are large-scale and aggregative in nature. With a number of studies falling somewhere in between these main trends, the conducted review challenges the conventional “qualitative-quantitative-mixed methods” distinction that has been traditionally used in reviews of previous research (e.g., Pike, 2002).

Texts were the dominant data type in the surveyed studies, and content analysis was the most frequently used analytical procedure in four out of five topic areas: service quality, destination image and reputation, experiences and behavior, and eWOM. A few studies did not use the term “content analysis” (e.g., Crotts, Mason, and Davis [2009] called their approach “stance-shift analysis”); however, by applying the principles of categorization and systematic data reduction to textual data, they adhered to Shapiro’s and Markoff’s (1997, p. 18) “minimal definition,” which states that content analysis is “any methodological measurement applied to text (or other symbolic material) for social science purposes.” Relatively rare alternatives to content analysis included discourse analysis (Azariah, 2012; Watson, Morgan, & Hemmington, 2008), narrative analysis (Berger & Greenspan, 2008; Bosangit, McCabe, & Hibbert, 2009), and thematic analysis (Ekiz, Khoo-Lattimore, &

Memarzadeh, 2012; Mkonzo, Markwell, & Wilson, 2013). A smaller number of studies applied content analysis to photos (e.g., Stylianou-Lambert, 2012) and videos (e.g., Reino & Hary, 2011). However, few studies have extended beyond obtaining category frequencies and have actually made statistical comparisons (for textual data: Crotts et al., 2009; Jiang, Gretzel, & Law, 2010; C. C. Lee & Hu, 2005; W. Lu & Stepchenkova, 2011; for visual data: Stepchenkova & Zhan, 2013; Syed-Ahmad, Musa, Kobas, & Murphy, 2013).

Quantitative methods were applied to UGC that, in addition to textual data, contained some sort of numerical data, such as measures of satisfaction (e.g., Jeong & Jeon, 2008), perceived helpfulness of hotel reviews (e.g., Black & Kelly, 2009), perceived usefulness of blog entries (e.g., Zehrer et al., 2011), or message influence in the form of “likes” on Facebook (Kwok & Yu, 2013). To analyze such types of data (coded as “rating scores” in Table 1), researchers used regression analysis (e.g., Ye et al., 2008), correlation (e.g., Stringam, Gerdes, & Vanleeuwen, 2010) and correspondence analyses (e.g., S. E. Kim & Lehto, 2012), parametric (e.g., Del Chiappa & Dall’Agilio, 2012) and nonparametric (e.g., Zehrer et al., 2011) analyses of means, and cluster (e.g., H. Li, Ye & Law, 2012) and component (e.g., Q. Ye, Li, Wang, & Law, 2012) analyses. Linear regression was the most frequently applied procedure (e.g., Jiang et al., 2010; Q. Lu, Xiao, & Ye, 2012; Otto, Sanford, & Chuang, 2009).

Several articles dealt with favorability of textual messages using computer-assisted algorithms. Sentiment analysis (other terms include favorability, valence, or polarity analysis), i.e., computational detection and study of opinions, emotions, and subjectivities in text, is a long-standing issue in content analysis (Krippendorff, 2004). The use of sentiment analysis and related approaches has gained popularity in recent years, given the advances in machine learning methods with respect to natural language processing and the availability of large data sets that are used to “train” machine learning algorithms (Duan, Cao, Yu, & Levy, 2013). In the surveyed articles, computational sentiment analysis was applied to provide a detailed analysis of consumer reviews of hotels (Duan et al., 2013; S. Liu, Law, Rong, Li, & Hall, 2013) and to explore the applicability of the technique in measuring traveler sentiment from tweets on the subject of Florida (Stepchenkova, Kirilenko, & Kim, 2013).

Recent technology and software advancements allow the extraction of “contextual information in addition to UGC data, i.e., timing, text tags, and geo-coordinates of photos or tweets, which expands the applicability scope of research involving UGC. Seven studies (Ardiizzone et al., 2012; Girardin et al., 2007, 2008; Kurashima et al., 2005; X. Lu et al., 2010; Popescu, Grefenstette, & Moellic, 2009; Y.-T. Zheng, Zha, & Chua, 2012) used a procedure named geo-visualization to provide spatial or temporal patterns of tourist flows. In the absence of established methodologies

for this type of research, researchers develop their own procedures by drawing on state-of-the-art advancements from the information sciences field. From a somewhat similar angle, a study by Pang, Hao, Yuan, Hu, Cai, and Zhang (2011) offered a model that “summarized” tourist destinations through both textual and visual descriptions found online. The authors designed a methodology for collecting destination-representative tags from travel blogs and then selected representative photos to visualize these tags. In all studies from the mobility patterns topic area, the procedures involved a high degree of formalization and the ability to handle large volumes of data.

Sampling is a major consideration in research involving UGC. While sampling methods for content analysis of general media have been established, tested, and validated (e.g., Lacy, Robinson, & Riffe, 1995; Riffe, Lacy, Nagovan, & Burkum, 1996), rigorous research with respect to UGC sampling is crucially needed. As Holsti (1969, p. 130) stated, sampling procedure “may involve as many as three steps: selecting *sources* of communication, *sampling* documents, and sampling *within documents*.” To give a research question a fair chance of being answered correctly, purposive sampling (Krippendorff, 2004, p. 113) is generally appropriate for selecting sources of UGC data. For the next two steps of Holsti’s procedure, however, a randomization algorithm is required, unless researchers take the whole data like, for example, all blogs about Charleston, South Carolina (Pan et al., 2007). Granted, the issues of sample representativeness and generalizability of results may seem less relevant when researchers utilize the big data aspect of UGC and collect all available data for a certain period and/or location. However, the completeness of big data that are openly available has been questioned, as “large data sets from Internet sources are of unreliable, prone to outages and losses, and these errors and gaps are magnified when multiple data sets are used together” (Boyd & Crawford, 2012, p. 668).

To complicate the picture, it is generally unknown who the producers of UGC are. Personal information of UGC creators is not always available because UGC platforms often grant anonymity rights to their members, who exercise these rights. Research that attempts to investigate demographic and psychographic profile of UGC creators reports results that differ widely (e.g., Stockl, Kosyak, Von Walter, & Hess, 2006; Gretzel, Yoo, & Purifoy, 2007). The situation when users of social media are different from general population, UGC creators are different from social media users, full UGC data is not easily accessible, and UGC creators tend not to provide their personal data create multiple layers of uncertainty for generalizability of research results obtained with the use of UGC. Overall, the review found that the questions of sample representativeness and generalizability of results from studied samples to populations of UGC materials, as well as discussions about what the collected data actually represent and how reliable they are, were rarely addressed in the surveyed studies.

What Specialized Software Have Been Used to Collect, Process, and Analyze UGC Data?

UGC as a form of big data is a “socio-technical phenomenon” (Boyd & Crawford, 2012, p. 662), the technical aspect of which addresses all issues pertaining to the handling of large data sets at both the data collection and data analysis stages of research. UGC is regularly distributed as single entries on consumer review websites and in blogs, or as discussion threads in virtual communities, a mode that does not lend itself well to aggregation and download. The “immediate” nature of UGC from such social media platforms as Facebook and Twitter and the limited capacity for searching through their archives have also been noted (Boyd & Crawford, 2012). Thus, locating and collecting UGC have been identified as one of the challenges that researchers face (Akehurst, 2009; Kline, Barreda, Gerdes, & Gregory, 2010). This review found that UGC data were collected “by hand” in most studies. A manual approach is time-consuming, laborious, and generally limits the sample size. The manual handling of data also decreases the degree of formalization in the collection process and, ultimately, facilitates the creeping of researcher’s bias into the data.

In the corpus of reviewed studies, 24, 38, and 18 studies employed specialized software for data collection, data analysis, or both, respectively. Programs for data collection were largely self-developed applications (SDAs) that are referred to as crawlers, spiders, robots, or worms. These applications “automate the process of visiting a group of sites, extract useful content from that site, and then store that information in a database for later retrieval” (Kline et al., 2010, n.p.). Web harvesting (i.e., collecting UGC information by robot software) is a controversial practice, since it adds extra traffic to a site and thus increases hosting costs, without bringing in the accompanying revenue (Johnson et al., 2012). In addition, the question of UGC ownership brings forth such issues as whether access to UGC needs to be negotiated with website owners prior to collecting it and whether or not unauthorized use of UGC is in fact copyright infringement (Jennings & Yates, 2009; Johnson et al., 2012). From the perspective of protection of human subjects (i.e., UGC creators), the attribution of cited UGC content can be established with powerful search engines and, potentially, traced to a real person.

Software for computer-assisted text analysis generally belonged to two classes. The first class are programs that support automatic coding of text using dictionaries and typically include word frequency analysis, category frequency counts, keyword-in-context, cluster analysis, and visualization functions. For example, C. C. Lee and Hu (2005) used the frequency and clustering functionalities of CATPAC to find the most popular keywords in e-complaints on hotel performance and to examine the associations among them. The quantification of attitudinal descriptors built into General Inquirer allowed Jiang et al. (2010) to quantify the negative sentiments and test the

attribution theory in the context of online hotel reviews. Other programs to address sentiment analysis included SENTIWORDNET (S. Liu et al., 2013) and the CLIPS package (Stepchenkova et al., 2013). The second class of software (e.g., ATLAS.ti, MAXQDA, NVivo) was used for operations associated with theme identification and category coding and served as “an electronic version of the set of marginal notes, cross-references and notepad jottings that a researcher will generate when analyzing a set of texts by hand” (Lowe, 2002, p. 1). For example, NVivo8 was used to facilitate the analysis of 200 web-based consumer complaints (Sparks & Browning, 2010).

The review found that, in general, the surveyed studies are vague with respect to how they collect UGC data and that the technical details of data analyses are often incomplete and that descriptions of the software functionalities are limited. The fact that a sizable portion of the software programs used in the surveyed studies were SDAs implies that having the skills to access, handle, and analyze this type of data introduces a new type of “digital divide” (Organisation for Economic Co-operation and Development, 2001), where those with computational and/or programming backgrounds often have an advantage (Boyd & Crawford, 2012; Manovich, 2011). An indication that this digital divide indeed exists is that the studies that utilized crawlers and web spiders for data search were published mainly in nontourism and hospitality conference proceedings (e.g., *Web Information Systems Engineering (WISE)* or *International Symposium on LBS and Telecartography*) and journals (e.g., *Journal of Services Marketing*). This observation has direct implications for educational institutions offering tourism and hospitality graduate programs.

CONCLUSION

The widespread use of social media combined with the developments in software instruments is drawing the attention of researchers toward selecting UGC as a research mode in tourism and hospitality applications. Presently, UGC research is mostly concerned with applications that explore the various aspects of service quality, destination image and reputation, experiences and behavior, the persuasive power of UGC as eWOM, as well as tourist mobility patterns. Textual UGC and content analysis are the leading data type and research method employed, respectively. The overall trend, however, is expansion of the repertoire of analytical and computational procedures, with an increasing number of studies using specialized software to manage the big data dimension of UGC. Researchers with a computational background have entered the field of tourism and hospitality applications and are applying developments from information sciences to speed up data collection, handling, and analysis.

This review contends that grand top-down theories make space for smaller-scale bottom-up projects (cf. Stone, 1997). Studies often do not

explicitly report the theoretical perspectives through which the analyses were conducted. This lack of theoretical foundations may stem from the immediacy of research applications and the view that clear-cut managerial implications may substitute for a sound theoretical framework. From the methodology angle, the possibilities of using UGC to statistically test theories have not yet been fully utilized. The issues of sampling representativeness, completeness of the data, reliability, validity, and generalizability of research results are often left out of discussion. The detected trends of researchers' interest are tourist mobility patterns, including visualization of digital footprints of tourists, sentiment analysis of consumer reviews and tourist experiences, business-to-consumer communications, as well as crisis-induced and other types of "in real-time" tourist behavior.

REFERENCES

- Aggett, M. (2011). Cruise passengers' complaints: An analysis of online reviews. In *Cruise Sector Challenges* (pp. 147-161). Retrieved from http://link.springer.com/chapter/10.1007/978-3-8349-6871-5_9
- Akehurst, G. (2009). User generated content: The use of blogs for tourism organisations and tourism consumers. *Service Business*, 3(1), 51-61.
- Ardizzone, E., Di Miceli, F., La Cascia, M., & Mazzola, G. (2012). Extracting touristic information from online image collections. In *Signal Image Technology and Internet Based Systems (SITIS), 2012 Eighth International Conference on 25-29 Nov. 2012* (pp. 482-488). Retrieved from http://ieeexplore.ieee.org/xpls/abs_all.jsp?arnumber=6395133
- Arsal, I., Woosnam, K. M., Baldwin, E. D., & Backman, S. J. (2010). Residents as travel destination information providers: An online community perspective. *Journal of Travel Research*, 49(4), 400-413.
- Au, N., Buhalis, D., & Law, R. (2009). Complaints on the online environment—the case of Hong Kong hotels. In *Information and Communication Technologies in Tourism 2009, Proceedings of the International Conference in Amsterdam, The Netherlands, 2009* (pp. 73-85). Amsterdam, The Netherlands: Springer-Verlag/Wien.
- Au, N., Law, R., & Buhalis, D. (2010). The impact of culture on eComplaints: Evidence from Chinese consumers in hospitality organisations. In *Information and Communication Technologies in Tourism 2010* (pp. 285-296). Vienna, Austria: Springer Vienna.
- Axup, J., & Viller, S. (2005). Augmenting travel gossip: Design for mobile communities. In *Proceedings of the 17th Australia conference on Computer-Human Interaction: Citizens Online: Considerations for Today and the Future* (pp. 1-4). Retrieved from <http://dl.acm.org/citation.cfm?id=1108436>
- Azariah, D. R. (2012). When travel meets tourism: Tracing discourse in Tony Wheeler's blog. *Critical Studies in Media Communication*, 29(4), 275-291.
- Banyai, M. (2010). Dracula's image in tourism: Western bloggers versus tour guides. *European Journal of Tourism Research*, 3(1), 5-22.

- Banyai, M. (2012). Travel blogs: A reflection of positioning strategies? *Journal of Hospitality Marketing & Management*, 21(4), 421–439.
- Banyai, M., & Glover, T. D. (2011). Evaluating research methods on travel blogs. *Journal of Travel Research*. Retrieved from <http://jtr.sagepub.com/content/early/2011/06/06/0047287511410323.abstract>
- Barrett, C. (1999, January 26). *Anatomy of a weblog site* [Web log post]. Retrieved from <http://camworld.org/archives/001177.html>
- Bazaar Voice. (2012). *Talking to strangers: Millennials trust people over brands*. Retrieved on from http://resources.bazaarvoice.com/rs/bazaarvoice/images/201202_Millennials_whitepaper.pdf
- Berger, I. E., & Greenspan, I. (2008). High (on) technology: Producing tourist identities through technologized adventure. *Journal of Sport & Tourism*, 13(2), 89–114.
- Black, H. G., & Kelley, S. W. (2009). A storytelling perspective on online customer reviews reporting service failure and recovery. *Journal of Travel & Tourism Marketing*, 26(2), 169–179.
- Bosangit, C., McCabe, S., & Hibbert, S. (2009). What is told in travel blogs? Exploring travel blogs for consumer narrative analysis. In *Information and Communication Technologies in Tourism 2009*, 61–71.
- Boyd, D., & Crawford, K. (2012). Critical questions for big data. *Information, Communication & Society*, 15(5), 662–679. doi:10.1080/1369118X.2012.678878
- Bulchand-Gidumal, J., Melián-González, S., & López-Valcárcel, B. G. (2011). Improving hotel ratings by offering free Wi-Fi. *Journal of Hospitality and Tourism Technology*, 2(3), 235–245.
- Capriello, A., Mason, P. R., Davis, B., & Crotts, J. C. (2011). Farm tourism experiences in travel reviews: A cross-comparison of three alternative methods for data analysis. *Journal of Business Research*. Retrieved from <http://www.sciencedirect.com/science/article/pii/S0148296311003274>
- Carson, D. (2008). Theblogosphere' as a market research tool for tourism destinations: A case study of Australia's Northern Territory. *Journal of Vacation Marketing*, 14(2), 111–119.
- Chaves, M., Gomes, R., & Pedron, C. (2012). Decision making based on Web 2.0 Data: The small and medium hotel management. Retrieved from <http://repositorio-cientifico.uatlantica.pt/handle/10884/639>
- Chen, H.-J., Chen, P.-J., & Wang, M.-H. (2012). Application of perception-promotion matrix model—the case of Kaohsiung City. *Journal of Travel & Tourism Marketing*, 29(2), 163–184.
- Choi, S., Lehto, X. Y., & Morrison, A. M. (2007). Destination image representation on the web: Content analysis of Macau travel related websites. *Tourism Management*, 28(1), 118–129.
- Cox, C., Burgess, S., Sellitto, C., & Buultjens, J. (2009). The role of user-generated content in tourists' travel planning behavior. *Journal of Hospitality Marketing & Management*, 18(8), 743–764.
- Crawford-Welch, S., & McCleary, K. W. (1992). An identification of the subject areas and research techniques used in five hospitality-related journals. *International Journal of Hospitality Management*, 11(2), 155–167.

- Crotts, J. C., Davis, B. H., & Mason, P. R. (2012). Analysing blog content for competitive advantage: Lessons learned in the application of software aided linguistics analysis. *Social Media in Travel, Tourism and Hospitality*, 281.
- Crotts, J. C., Mason, P. R., & Davis, B. (2009). Measuring guest satisfaction and competitive position in the hospitality and tourism industry. *Journal of Travel Research*, 48(2), 139–151.
- Day, J., Skidmore, S., & Koller, T. (2002). Image selection in destination positioning: A new approach. *Journal of Vacation Marketing*, 8(2), 177–186.
- De Ascaniis, S., Bischof, N., & Cantoni, L. (2013). Building destination image through online opinionated discourses. The case of Swiss mountain destinations. In *Information and Communication Technologies in Tourism 2013* (pp. 94–106). Retrieved from http://link.springer.com/chapter/10.1007/978-3-642-36309-2_9
- Del Chiappa, G., & Dall'Aglio, S. (2012). Factors influencing travellers'e-ratings and e-complaints about hotel services: Insights from an Italian tourism destination. In *Information and Communication Technologies in Tourism 2012* (pp. 448–459). Retrieved from http://link.springer.com/chapter/10.1007/978-3-7091-1142-0_39
- Duan, W., Cao, Q., Yu, Y., & Levy, S. (2013). Mining online user-generated content: Using sentiment analysis technique to study hotel service quality. In *System Sciences (HICSS), 2013 46th Hawaii International Conference on* (pp. 3119–3128). Retrieved from http://ieeexplore.ieee.org/xpls/abs_all.jsp?arnumber=6480220
- Duverger, P. (2013). Curvilinear effects of user-generated content on hotels' market share: A dynamic panel-data analysis. *Journal of Travel Research*. Retrieved from <http://jtr.sagepub.com/content/early/2013/02/28/0047287513478498.abstract>
- Dwivedi, M. (2009). Online destination image of India: A consumer based perspective. *International Journal of Contemporary Hospitality Management*, 21(2), 226–232.
- Dwivedi, M., Yadav, A., & Patel, V. R. (2009). The online destination image of Goa. *Worldwide Hospitality and Tourism Themes*, 1(1), 25–39.
- Dwyer, L., Gill, A., & Seetaram, N. (2012). *Handbook of research methods in tourism: Quantitative and qualitative approaches*. Retrieved from http://books.google.com/books?hl=en&lr=&id=qTwb_VH7qIoC&oi=fnd&pg=PR3&dq=handbook+of+research+methods+in+tourism+larry+dwyer&ots=q-ZFOb0Ipl&sig=pVieYNzBZPwxCPtx5bVdT1lElws
- Ekiz, E., Khoo-Lattimore, C., & Memarzadeh, F. (2012). Air the anger: Investigating online complaints on luxury hotels. *Journal of Hospitality and Tourism Technology*, 3(2), 96–106.
- Enoch, Y., & Grossman, R. (2010). Blogs of Israeli and Danish backpackers to India. *Annals of Tourism Research*, 37(2), 520–536.
- García-Barriocanal, E., Sicilia, M. A., & Korfiatis, N. (2010). Exploring hotel service quality experience indicators in user-generated content: A case using Tripadvisor data. Retrieved from <http://aisel.aisnet.org/mcis2010/33/>
- Giannakos, M. N., Pateli, A., & Pappas, I. (2012). Investigation of the hotel customers' perceptions: A study based on user-generated content of online booking platforms. In *Proc. of 2nd Advances in Hospitality and Tourism Marketing &*

- Management Conference (AHTMMC'12)*. Retrieved from http://www.ionio.gr/~mgiannak/AHTMMC_2011.pdf
- Girardin, F., Calabrese, F., Fiore, F. D., Ratti, C., & Blat, J. (2008). Digital footprinting: Uncovering tourists with user-generated content. *Pervasive Computing, IEEE*, 7(4), 36–43.
- Girardin, F., Dal Fiore, F., Blat, J., & Ratti, C. (2007). Understanding of tourist dynamics from explicitly disclosed location information. Retrieved from http://senseable.mit.edu/papers/pdf/2007_Girardin_et_al_Tourist_dynamics_LBS-TC.pdf
- Goodchild, M. F. (2007). Citizens as voluntary sensors: Spatial data infrastructure in the world of web 2.0. *International Journal of Spatial Data Infrastructures Research*, 2, 24–32.
- Goulding, C., Saren, M., & Lindridge, A. (2013). Reading the body at von Hagen's "body worlds." *Annals of Tourism Research*, 40, 306–330.
- Gretzel, U., & Yoo, K. H. (2008). Use and impact of online travel reviews. In *Information and Communication Technologies in Tourism 2008* (pp. 35–46). Retrieved from http://link.springer.com/chapter/10.1007%2F978-3-211-77280-5_4?LI=true
- Gretzel, U., Yoo, K. H., & Purifoy, M. (2007, February). *Online travel review study: Role & impact of online travel reviews*. Retrieved from <http://www.tripadvisor.com/pdfs/OnlineTravelReviewReport.pdf>
- Gu, B., & Ye, Q. (2010). The impact of online service recovery on customer satisfaction: Empirical evidences from service operations in China. Retrieved from http://papers.ssrn.com/sol3/papers.cfm?abstract_id=1569946
- Harrison-Walker, L. J. (2001). E-complaining: A content analysis of an Internet complaint forum. *Journal of Services Marketing*, 15(5), 397–412.
- Hays, S., Page, S. J., & Buhalis, D. (2012). Social media as a destination marketing tool: Its use by national tourism organisations. Retrieved from <http://www.tandfonline.com/doi/abs/10.1080/13683500.2012.662215>
- Hjalager, A.-M., & Nordin, S. (2011). User-driven innovation in tourism—a review of methodologies. *Journal of Quality Assurance in Hospitality & Tourism*, 12(4), 289–315.
- Holsti, O. R. (1969). *Content analysis for the social sciences and humanities*. Retrieved from <http://library.wur.nl/WebQuery/clc/410414>
- Hvass, K. A., & Munar, A. M. (2012). The takeoff of social media in tourism. *Journal of Vacation Marketing*, 18(2), 93–103.
- Inversini, A., Marchiori, E., Dedekind, C., & Cantoni, L. (2010). Applying a conceptual framework to analyze online reputation of tourism destinations. In *Information and Communication Technologies in Tourism 2010* (pp. 321–332). Vienna, Austria: Springer Vienna.
- Ip, C., Cheung, C., Law, R., & Au, N. (2011). Travel preferences of overseas destinations by mainland Chinese online users. In *Information and Communication Technologies in Tourism 2011* (pp. 139–150). Retrieved from http://link.springer.com/chapter/10.1007/978-3-7091-0503-0_12
- Jamal, T., Smith, B., & Watson, E. (2008). Ranking, rating and scoring of tourism journals: Interdisciplinary challenges and innovations. *Tourism Management*, 29(1), 66–78.

- Jani, D., & Hwang, Y.-H. (2011). User-generated destination image through weblogs: A comparison of pre-and post-visit images. *Asia Pacific Journal of Tourism Research*, 16(3), 339–356.
- Jani, D., Jeong, H. K., & Hwang, Y.-H. (2011). Analyzing international tourists' functional information needs: A comparative analysis of inquiries in an on-line travel forum. *Journal of Business Research*. Retrieved from <http://www.sciencedirect.com/science/article/pii/S0148296311003158>
- Jasco, P. (2008). Testing the calculation of a realistic h-index in Google Scholar, Scopus, and Web of Science for FW Lancaster. Retrieved from <http://www.ideals.illinois.edu/handle/2142/9493>
- Jennings, F., & Yates, J. (2009). Scrapping over data: Are the data scrapers' days numbered? *Journal of Intellectual Property Law & Practice*, 4(2), 120–129.
- Jeong, M., & Jeon, M. M. (2008). Customer reviews of hotel experiences through consumer generated media (CGM). *Journal of Hospitality & Leisure Marketing*, 17(1/2), 121–138.
- Jiang, J., Gretzel, U., & Law, R. (2010). Do negative experiences always lead to dissatisfaction? Testing attribution theory in the context of online travel reviews. In *Information and Communication Technologies in Tourism 2010* (pp. 297–308). Vienna, Austria: Springer Vienna.
- Johnson, P. A., Sieber, R. E., Magnien, N., & Ariwi, J. (2012). Automated web harvesting to collect and analyse user-generated content for tourism. *Current Issues in Tourism*, 15(3), 293–299.
- Kim, J., & Hardin, A. (2010). The impact of virtual worlds on word-of-mouth: Improving social networking and servicescape in the hospitality industry. *Journal of Hospitality Marketing & Management*, 19(7), 735–753.
- Kim, S. E., & Lehto, X. Y. (2012). The voice of tourists with mobility disabilities: Insights from online customer complaint websites. *International Journal of Contemporary Hospitality Management*, 24(3), 451–476.
- Klassen, T. P., Jadad, A. R., & Moher, D. (1998). Guides for reading and interpreting systematic reviews: I. Getting started. *Archives of Pediatrics & Adolescent Medicine*, 152(7), 700.
- Kline, S., Barreda, A., Gerdes, J., & Gregory, S. (2010). *Visual interpretations of guests' hotel stay experience: A case study*. EuroCHRIE. Retrieved from <http://www.eurochrie2010.nl/publications/1038.pdf>
- KnowWPC. (2012, May 8) *eBay study: How to build trust and improve the shopping experience*. Retrieved from <http://knowwpcarey.com/article.cfm?cid=25&aid=1171>
- Krippendorff, K. (2004). *Content analysis: An introduction to its methodology*. Retrieved from http://books.google.com/books?hl=en&lr=&id=q657o3M3C8cC&oi=fnd&pg=PA3&dq=2002+Content+analysis:+An+introduction+to+its+methodology,+2nd+ed.+Thousand+Oaks:+Sage+Publications,+Inc.&ots=bKcfu_M7EZ&sig=0MzG3vNKf3hyeLEDqXW8tOQvMHU
- Kurashima, T., Tezuka, T., & Tanaka, K. (2005). Blog map of experiences: Extracting and geographically mapping visitor experiences from urban blogs. In A. H. H. Ngu, M. Kitsuregawa, E. J. Neuhold, J.-Y. Chung, & Q. Z. Sheng (Eds.), *Web Information Systems Engineering–WISE 2005* (pp. 496–503). Berlin, Germany: Springer Berlin Heidelberg.

- Kwok, L., & Yu, B. (2013). Spreading social media messages on Facebook: An analysis of restaurant business-to-consumer communications. *Cornell Hospitality Quarterly*, 54(1), 84–94.
- Lacy, S., Robinson, K., & Riffe, D. (1995). Sample size in content analysis of weekly newspapers. *Journalism & Mass Communication Quarterly*, 72(2), 336–345.
- Law, R., & Cheung, S. (2010). The perceived destination image of Hong Kong as revealed in the travel blogs of mainland Chinese tourists. *International Journal of Hospitality & Tourism Administration*, 11(4), 303–327.
- Lee, C. C., & Hu, C. (2005). Analyzing hotel customers' e-complaints from an Internet complaint forum. *Journal of Travel & Tourism Marketing*, 17(2/3), 167–181.
- Lee, H., Law, R., & Murphy, J. (2011). Helpful reviewers in TripAdvisor, an online travel community. *Journal of Travel & Tourism Marketing*, 28(7), 675–688.
- Lehto, X., Park, J., Park, O., & Lehto, M. (2007). Text analysis of consumer reviews: The case of virtual travel firms. In M. J. Smith & G. Salvendy (Eds.), *Human interface and the management of information. Methods, techniques and tools in information design* (pp. 490–499). Berlin, Germany: Springer-Verlag Berlin Heidelberg.
- Leung, D., Law, R., & Lee, H. A. (2011). The perceived destination image of Hong Kong on Ctrip. com. *International Journal of Tourism Research*, 13(2), 124–140.
- Leung, D., Law, R., van Hoof, H., & Buhalis, D. (2013). Social media in tourism and hospitality: A literature review. *Journal of Travel & Tourism Marketing*, 30(1/2), 3–22.
- Leung, R., Schuckert, M., & Yeung, E. (2013). Attracting user social media engagement: A study of three budget airlines Facebook pages. In *Information and communication technologies in tourism 2013* (pp. 195–206). Retrieved from http://link.springer.com/chapter/10.1007/978-3-642-36309-2_17
- Levy, S. E., Duan, W., & Boo, S. (2013). An analysis of one-star online reviews and responses in the Washington, DC, lodging market. *Cornell Hospitality Quarterly*, 54(1), 49–63.
- Li, G., Law, R., Vu, H. Q., & Rong, J. (2012). Discovering the hotel selection preferences of Hong Kong inbound travelers using the Choquet Integral. *Tourism Management*. Retrieved from <http://www.sciencedirect.com/science/article/pii/S0261517712001975>
- Li, H., Ye, Q., & Law, R. (2012). Determinants of customer satisfaction in the hotel industry: An application of online review analysis. Retrieved from <http://www.tandfonline.com/doi/abs/10.1080/10941665.2012.708351>
- Li, N., Buhalis, D., & Zhang, L. (2013). Interdisciplinary research on information science and tourism. In *Information and communication technologies in tourism 2013* (pp. 302–313). Retrieved from http://link.springer.com/chapter/10.1007/978-3-642-36309-2_26
- Li, X., & Wang, Y. (2011). Depicting image of China as a tourism destination: A travel blog approach. Retrieved from http://scholarworks.umass.edu/gradconf_hospitality/2011/Presentation/91/
- Lim, Y., Chung, Y., & Weaver, P. A. (2012). The impact of social media on destination branding Consumer-generated videos versus destination marketer-generated videos. *Journal of Vacation Marketing*, 18(3), 197–206.

- Lin, Y. S., & Huang, J. Y. (2006). Internet blogs as a tourism marketing medium: A case study. *Journal of Business Research*, 59(10), 1201–1205.
- Liu, Y. (2012). The influence factors to the enterprise microblogs: A research of the restaurant enterprise microblogs on Weibo.com. In *Computing Technology and Information Management (ICCM), 2012 8th International Conference on Apr. 24–26, 2012* (Vol. 2, pp. 590–595). Retrieved from http://ieeexplore.ieee.org/xpls/abs_all.jsp?arnumber=6268568
- Liu, S., Law, R., Rong, J., Li, G., & Hall, J. (2013). Analyzing changes in hotel customers' expectations by trip mode. *International Journal of Hospitality Management*, 34, 359–371. Retrieved from <http://www.sciencedirect.com/science/article/pii/S0278431912001545>
- Lowe, W. (2002). Software for content analysis—a review. Cambridge, MA: Weatherhead Center for International Affairs and the Harvard Identity Project.
- Lu, Q., Xiao, L., & Ye, Q. (2012). Investigating the impact of online word-of-mouth on hotel sales with panel data. In *Management Science and Engineering (ICMSE), 2012 International Conference on Sep. 20–22, 2012* (pp. 3–9). Retrieved from http://ieeexplore.ieee.org/xpls/abs_all.jsp?arnumber=6414153
- Lu, W., & Stepchenkova, S. (2011). Ecotourism experiences reported online: Classification of satisfaction attributes. *Tourism Management*. Retrieved from <http://www.sciencedirect.com/science/article/pii/S0261517711001567>
- Lu, X., Wang, C., Yang, J. M., Pang, Y., & Zhang, L. (2010). Photo2Trip: Generating travel routes from geo-tagged photos for trip planning. *Proceedings of the international conference on multimedia* (pp. 143–152). Retrieved from <http://dl.acm.org/citation.cfm?id=1873972>
- Lu, Z., & Feng, Z. (2006). The post-purchase evaluation of the China's star rated hotel service quality: A content analysis of guest comments on third party hotel booking websites. *The Journal of China Tourism Research*, 2(1/2), 93–103.
- Maansson, M. (2011). Mediatized tourism. *Annals of Tourism Research*, 38(4), 1634–1652.
- Magnini, V. P., Crotts, J. C., & Zehrer, A. (2011). Understanding customer delight: An application of travel blog analysis. *Journal of Travel Research*, 50(5), 535–545.
- Magnini, V. P., Kara, D., Crotts, J. C., & Zehrer, A. (2012). Culture and service-related positive disconfirmations An application of travel blog analysis. *Journal of Vacation Marketing*, 18(3), 251–257.
- Manovich, L. (2011). Trending: The promises and the challenges of big social data. *Debates in the Digital Humanities*. Retrieved from http://www.manovich.net/DOCS/Manovich_trending_paper.pdf
- Marchiori, E., Eynard, D., Inversini, A., Cantoni, L., & Cerretti, F. (2011). Harvesting online contents: An analysis of hotel reviews websites. In *Proceedings of the information and communication technologies in tourism* (pp. 101–112). Retrieved from www.academia.edu/download/30501502/07_harvesting_online_contents_an_analysis_of_hotel_reviews_websites.pdf
- Maslov, S., & Redner, S. (2008). Promise and pitfalls of extending Google's PageRank algorithm to citation networks. *The Journal of Neuroscience*, 28(44), 11103–11105.
- Maurer, C., & Schaich, S. (2011). Online customer reviews used as complaint management tool. In *Information and communication technologies in tourism* (pp.

- 499–511). Retrieved from http://link.springer.com/chapter/10.1007/978-3-7091-0503-0_40
- McKenna, R. (1991). *Relationship marketing: Successful strategies for the age of the customer*. Reading, MA: Addison-Wesley.
- McKercher, B., Law, R., & Lam, T. (2006). Rating tourism and hospitality journals. *Tourism Management*, 27(6), 1235–1252.
- Miles, M. B., & Huberman, A. M. (1994). *Qualitative data analysis: An expanded sourcebook*. Retrieved from [http://books.google.com/books?hl=en&lr=&id=U4IU_-wJ5QEC&oi=fnd&pg=PR12&dq=Miles,+M.+B.,+%26+Huberman,+A.+M.+\(1994\).+Qualitative+data+analysis:+An+expanded+sourcebook+\(2nd+ed.\).+&ots=kCWB2KSY1W&sig=ZSponHfjllyoNMNGPH4czgLeu4](http://books.google.com/books?hl=en&lr=&id=U4IU_-wJ5QEC&oi=fnd&pg=PR12&dq=Miles,+M.+B.,+%26+Huberman,+A.+M.+(1994).+Qualitative+data+analysis:+An+expanded+sourcebook+(2nd+ed.).+&ots=kCWB2KSY1W&sig=ZSponHfjllyoNMNGPH4czgLeu4)
- Mkono, M. (2013). African and Western tourists: Object authenticity quest? *Annals of Tourism Research*, 41, 195–214.
- Mkono, M., Markwell, K., & Wilson, E. (2013). Applying Quan and Wang's structural model of the tourist experience: A Zimbabwean netnography of food tourism. *Tourism Management Perspectives*, 5, 68–74.
- Murakami, K. H., Kawamura, H., & Suzuki, K. (2012a). An analysis of word of mouth in travel blogs: Finding popular bloggers and blog entries. Retrieved from <http://www.apiems.net/conf2012/M4D3.pdf>
- Murakami, K. H., Kawamura, H., & Suzuki, K. (2012b). Earthquake's influence on inbound tourism: voices from the travel blogs. *Sustainable Tourism V: WIT Transactions on Ecology and The Environment*, 161, 43–53.
- Nimrod, G. (2012). Online communities as a resource in older adults' tourism. *The Journal of Community Informatics*, 8(1). Retrieved from <https://lib.njnu.edu.cn/proxy/Enlinkwkkoxe00x0v2upjmqxms1qzk/index.php/ciej/article/view/757>
- O'Connor, P. (2010). Managing a hotel's image on TripAdvisor. *Journal of Hospitality Marketing & Management*, 19(7), 754–772.
- O'Connor, P. (2011). An analysis of the use of Facebook by international hotel chains. Retrieved from http://scholarworks.umass.edu/refereed/ICHRIE_2011/Wednesday/9/
- Ong, C.-E., & du Cros, H. (2012). The post-Mao gazes: Chinese backpackers in Macau. *Annals of Tourism Research*, 39(2), 735–754.
- Organisation for Economic Co-operation and Development. (2001). *Understanding the digital divide*. Retrieved from <http://www.oecd.org/dataoecd/38/57/1888451.pdf>
- Organisation for Economic Co-operation and Development. (2007, April 12). *Participative web: User-generated content*. Retrieved from <http://www.oecd.org/dataoecd/57/14/38393115.pdf>
- Otto, J., Sanford, D., & Chuang, M. (2009). Using hotel web ratings data: Understanding customers' overall satisfaction ratings. *Tourism Review International*, 13(1), 51–63.
- Ögüt, H., & Onur Tacs, B. K. (2012). The influence of Internet customer reviews on the online sales and prices in hotel industry. *The Service Industries Journal*, 32(2), 197–214.
- Pan, B., MacLaurin, T., & Crotts, J. C. (2007). Travel blogs and the implications for destination marketing. *Journal of Travel Research*, 46(1), 35–45.

- Pang, Y., Hao, Q., Yuan, Y., Hu, T., Cai, R., & Zhang, L. (2011). Summarizing tourist destinations by mining user-generated travelogues and photos. *Computer Vision and Image Understanding*, 115(3), 352–363.
- Pantelidis, I. S. (2010). Electronic meal experience: A content analysis of online restaurant comments. *Cornell Hospitality Quarterly*, 51(4), 483–491.
- Paris, C. M. (2011). Understanding the statusphere and blogosphere: An analysis of virtual backpacker spaces. In *Information and Communication Technologies in Tourism 2011* (pp. 443–455). Retrieved from http://link.springer.com/chapter/10.1007/978-3-7091-0503-0_36
- Paris, C. M., & Rubin, S. (2013). Backpacking, social media, and crises: A discussion of online social convergence. In *Information and Communication Technologies in Tourism 2013* (pp. 207–217). Retrieved from http://link.springer.com/chapter/10.1007/978-3-642-36309-2_18
- Park, S.-Y., & Allen, J. P. (2013). Responding to online reviews: Problem solving and engagement in hotels. *Cornell Hospitality Quarterly*, 54(1), 64–73.
- Pearce, P. L. (2012). Tourists' written reactions to poverty in Southern Africa. *Journal of Travel Research*, 51(2), 154–165.
- Pechlaner, H., Zehrer, A., Matzler, K., & Abfalter, D. (2004). A ranking of international tourism and hospitality journals. *Journal of Travel Research*, 42(4), 328–332.
- Pengiran-Kahar, D. I. N. F., Syed-Ahmad, S. F., Ismail, S. H. S., & Murphy, J. (2010). Shared Arabian Muslim travel photos. In *Information and Communication Technologies in Tourism 2010* (pp. 543–554). Retrieved from http://link.springer.com/chapter/10.1007/978-3-211-99407-8_45
- Pike, S. (2002). Destination image analysis—a review of 142 papers from 1973 to 2000. *Tourism Management*, 23(5), 541–549.
- Popescu, A., Grefenstette, G., & Moëllic, P.-A. (2009). Mining tourist information from user-supplied collections. In *Proceedings of the 18th ACM conference on Information and knowledge management* (pp. 1713–1716). Retrieved from <http://dl.acm.org/citation.cfm?id=1646211>
- Racherla, P., & King, R. (2012). What we know and don't know about online word-of-mouth: A review and synthesis of the literature. Retrieved from http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2187040
- Ramanathan, U., & Ramanathan, R. (2013). Investigating the impact of resource capabilities on customer loyalty: A structural equation approach for the UK hotels using online ratings. *Journal of Services Marketing*, 27(5), 404–415.
- Reino, S., & Hay, B. (2011). The use of YouTube as a tourism marketing tool. In *Proceedings of the 42nd Annual Travel & Tourism Research Association Conference, London, Ontario, Canada* (Vol. 42). Retrieved from <http://ereseach.qmu.ac.uk/2315/>
- Rheingold, H. (1993). *The virtual community: Finding connection in a computerized world*. Retrieved from <http://dl.acm.org/citation.cfm?id=529182>
- Riffe, D., Lacy, S., Nagovan, J., & Burkum, L. (1996). The effectiveness of simple and stratified random sampling in broadcast news content analysis. *Journalism & Mass Communication Quarterly*, 73(1), 159–168.
- SAGE. (n.d.). *Help readers find your article: The importance of search engines*. Retrieved from <http://www.sagepub.com/journalgateway/findArticle.htm>

- Sarı, F. Ö., Alikılıç, Ö., & Onat, F. (2013). E-complaining: Analysis of lodging customers'e-complaints from a Turkish Internet website. Retrieved from http://www.atlantis-press.com/php/download_paper.php?id=6147
- Schmallegger, D., & Carson, D. (2009). Destination image projection on consumer-generated content websites: A case study of the flinders ranges. *Information Technology & Tourism*, 11(2), 111–127. doi:10.3727/109830509789994838
- Shakeela, A., & Weaver, D. (2012). Resident reactions to a tourism incident: Mapping a Maldivian Emoscape. *Annals of Tourism Research*, 39(3), 1337–1358.
- Shao, G. (2009). Understanding the appeal of user-generated media: A uses and gratification perspective. *Internet Research*, 19(1), 7–25.
- Shapiro, G., & Markoff, J. (1997). A matter of definition. In C. W. Roberts (Ed.), *Text analysis for the social sciences: Methods for drawing statistical inferences from texts and transcripts* (pp. 9–31). Mahwah, NJ: Lawrence Erlbaum Associates.
- Sigala, M. (2012). Social networks and customer involvement in new service development (NSD): The case of <http://www.mystarbucksidea.com>. *International Journal of Contemporary Hospitality Management*, 24(7), 966–990.
- Soderlund, M. (1998). Customer satisfaction and its consequences on customer behaviour revisited: The impact of different levels of satisfaction on word-of-mouth, feedback to the supplier and loyalty. *International Journal of Service Industry Management*, 9(2), 169–188.
- Son, A. (2011). International tourists' image of Zhangjiajie, China: Content analysis of travel blogs. *International Journal of Culture, Tourism and Hospitality Research*, 5(3), 306–315.
- Sparks, B. A., & Browning, V. (2010). Complaining in cyberspace: the motives and forms of hotel guests' complaints online. *Journal of Hospitality Marketing & Management*, 19(7), 797–818.
- Stepchenkova, S., Kirilenko, A., & Kim, H. (2013). Grassroots branding with Twitter: Amazing Florida. In *Information and Communication Technologies in Tourism 2013* (pp. 144–156). Retrieved from http://link.springer.com/chapter/10.1007/978-3-642-36309-2_13
- Stepchenkova, S., & Mills, J. E. (2010). Destination image: A meta-analysis of 2000–2007 research. *Journal of Hospitality Marketing and Management*, 19(6), 575–609.
- Stepchenkova, S., & Zhan, F. (2013). Visual destination images of Peru: Comparative content analysis of DMO and user-generated photography. *Tourism Management*, 36, 590–601. Retrieved from <http://www.sciencedirect.com/science/article/pii/S0261517712001422>
- Stöckl, R., Kosyak, A., Von Walter, B., & Hess, T. (2006). Success factors of communities for user driven content: The case of Ciao.com. *Proceedings of 12th Americas Conference on Information Systems* (pp. 4444–4452). Retrieved from <http://www.virtual-communities.net/mediawiki/images/1/15/VIRTCOM-1111.pdf>
- Stone, P. J. (1997). Thematic text analysis: New agendas for analyzing text content. In C. W. Roberts (Ed.), *Text analysis for the social sciences: Methods for drawing statistical inferences from texts and transcripts* (pp. 35–54). Mahwah, NJ: Lawrence Erlbaum Associates.

- Stringam, B. B., & Gerdes, J., Jr. (2010). An analysis of word-of-mouth ratings and guest comments of online hotel distribution sites. *Journal of Hospitality Marketing & Management*, 19(7), 773–796.
- Stringam, B. B., Gerdes, J., Jr., & Vanleeuwen, D. M. (2010). Assessing the importance and relationships of ratings on user-generated traveler reviews. *Journal of Quality Assurance in Hospitality & Tourism*, 11(2), 73–92.
- Stylianou-Lambert, T. (2012). Tourists with cameras: Reproducing or producing? *Annals of Tourism Research*, 39(4), 1817–1838.
- Syed-Ahmad, S. F., Musa, G., Klobas, J. E., & Murphy, J. (2013). Audience response to travel photos and Arab destination image. *Journal of Travel & Tourism Marketing*, 30(1/2), 161–164.
- Theng Loo, P., Boo, H. C., & Khoo-Lattimore, C. (2013). Profiling service failure and customer online complaint motives in the case of single failure and double deviation. *Journal of Hospitality Marketing & Management*. Retrieved from <http://www.tandfonline.com/doi/abs/10.1080/19368623.2013.724373>
- Tse, S. M., & Zhang, E. Y. (2012). Analysis of blogs and microblogs: A case study of Chinese bloggers sharing their Hong Kong Travel experiences. Retrieved from <http://www.tandfonline.com/doi/abs/10.1080/10941665.2012.658413>
- Tussyadiah, I. P., & Fesenmaier, D. R. (2009). Mediating tourist experiences: Access to places via shared videos. *Annals of Tourism Research*, 36(1), 24–40.
- Tussyadiah, I., & Zach, F. (2013). Social media strategy and capacity for consumer co-creation among destination marketing organizations. In *Information and communication technologies in tourism 2013* (pp. 242–253). Retrieved from http://link.springer.com/chapter/10.1007/978-3-642-36309-2_21
- Tuzovic, S. (2010). Frequent (flier) frustration and the dark side of word-of-web: Exploring online dysfunctional behavior in online feedback forums. *Journal of Services Marketing*, 24(6), 446–457.
- Twitter. (n.d.). *About Twitter*. Retrieved January 26, 2014, from <https://about.twitter.com/>
- van Aalst, J. (2010). Using Google Scholar to estimate the impact of journal articles in education. *Educational Researcher*, 39(5), 387–400.
- Wang, Y., & Fesenmaier, D. R. (2007). Collaborative destination marketing: A case study of Elkhart county, Indiana. *Tourism Management*, 28, 863–875.
- Wang, D., Park, S., & Fesenmaier, D. R. (2012). The role of smartphones in mediating the touristic experience. *Journal of Travel Research*, 51(4), 371–387.
- Watson, P., Morgan, M., & Hemmington, N. (2008). Online communities and the sharing of extraordinary restaurant experiences. *Journal of Foodservice*, 19(6), 289–302.
- Weed, M. (2006). Sports tourism research 2000–2004: A systematic review of knowledge and a meta-evaluation of methods. *Journal of Sport & Tourism*, 11(1), 5–30.
- Wenger, A. (2008). Analysis of travel bloggers' characteristics and their communication about Austria as a tourism destination. *Journal of Vacation Marketing*, 14(2), 169–176.
- Woodside, A. G., Cruickshank, B. F., & Dehuang, N. (2007). Stories visitors tell about Italian cities as destination icons. *Tourism Management*, 28(1), 162–174.

- Xiang, Z., & Gretzel, U. (2010). Role of social media in online travel information search. *Tourism Management*, 31(2), 179–188.
- Ye, Q., Gu, B., Chen, W., & Law, R. (2008). Measuring the value of managerial responses to online reviews: A natural experiment of two online travel agencies. In *ICIS 2008 Proceedings* (Paper 115). Retrieved from <http://aisel.aisnet.org/icis2008/115>
- Ye, Q., Law, R., Gu, B., & Chen, W. (2011). The influence of user-generated content on traveler behavior: An empirical investigation on the effects of e-word-of-mouth to hotel online bookings. *Computers in Human Behavior*, 27(2), 634–639.
- Ye, Q., Li, H., Wang, Z., & Law, R. (2012). The influence of hotel price on perceived service quality and value in e-tourism: An empirical investigation based on online traveler reviews. *Journal of Hospitality & Tourism Research*. Retrieved from <http://jht.sagepub.com/content/early/2012/04/16/1096348012442540.abstract>
- Yoo, K. H., & Gretzel, U. (2009). Comparison of deceptive and truthful travel reviews. In *Information and communication technologies in tourism 2009* (pp. 37–47). Retrieved from http://link.springer.com/chapter/10.1007/978-3-211-93971-0_4
- Zehrer, A., Crofts, J. C., & Magnini, V. P. (2011). The perceived usefulness of blog postings: An extension of the expectancy-disconfirmation paradigm. *Tourism Management*, 32(1), 106–113.
- Zhang, J. J., & Mao, Z. (2012). Image of all hotel scales on travel blogs: Its impact on customer loyalty. *Journal of Hospitality Marketing & Management*, 21(2), 113–131.
- Zhang, J. Q., Craciun, G., & Shin, D. (2010). When does electronic word-of-mouth matter? A study of consumer product reviews. *Journal of Business Research*, 63(12), 1336–1341.
- Zhang, Z., Ye, Q., & Law, R. (2011). Determinants of hotel room price: An exploration of travelers' hierarchy of accommodation needs. *International Journal of Contemporary Hospitality Management*, 23(7), 972–981.
- Zhang, Z., Ye, Q., Law, R., & Li, Y. (2010). The impact of e-word-of-mouth on the online popularity of restaurants: A comparison of consumer reviews and editor reviews. *International Journal of Hospitality Management*, 29(4), 694–700.
- Zhang, Z., Ye, Q., Song, H., & Liu, T. (2013). The structure of customer satisfaction with cruise-line services: An empirical investigation based on online word of mouth. *Current Issues in Tourism*. Advance online publication. doi:10.1080/13683500.2013.776020
- Zheng, T., Youn, H., & Kincaid, C. S. (2009). An analysis of customers' e-complaints for luxury resort properties. *Journal of Hospitality Marketing & Management*, 18(7), 718–729.
- Zheng, Y.-T., Zha, Z.-J., & Chua, T.-S. (2012). Mining travel patterns from geotagged photos. *ACM Transactions on Intelligent Systems and Technology*, 3(3), 56.